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Palaeomagnetism of Cretaceous to Tertiary sediments from southern Tibet: evidence for the extent of the northern margin of India prior to the collision with Eurasia

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Abstract

Palaeomagnetic studies have been carried out on mid-Cretaceous to latest Palaeocene sediments from the Tethyan Himalaya on the northern margin of the Indian Plate near the localities of Gamba (28.3°N, 88.5°E) and Duela (28.0°N, 89.2°E) in southern Tibet. A total of 61 sites (more than 700 cores) have been sampled mainly from limestones (Zongshan Formation, Jidula Formation Member II, Zongpu Formation), sandstones (Jidula Formation Member III) and graywackes (Gamba Group Member II), with ages well defined from detailed lithological and biostratigraphic analyses. Carriers of characteristic remanences are magnetite in limestones and graywackes, and hematite in sandstones. Reliable palaeodirections were obtained from 35 sites (356 specimens). Positive fold tests, antiparallel normal and reverse polarities, a northward trend of the palaeolatitudes with decreasing age, and absence of any metamorphic overprint indicate a primary origin for the characteristic remanences.

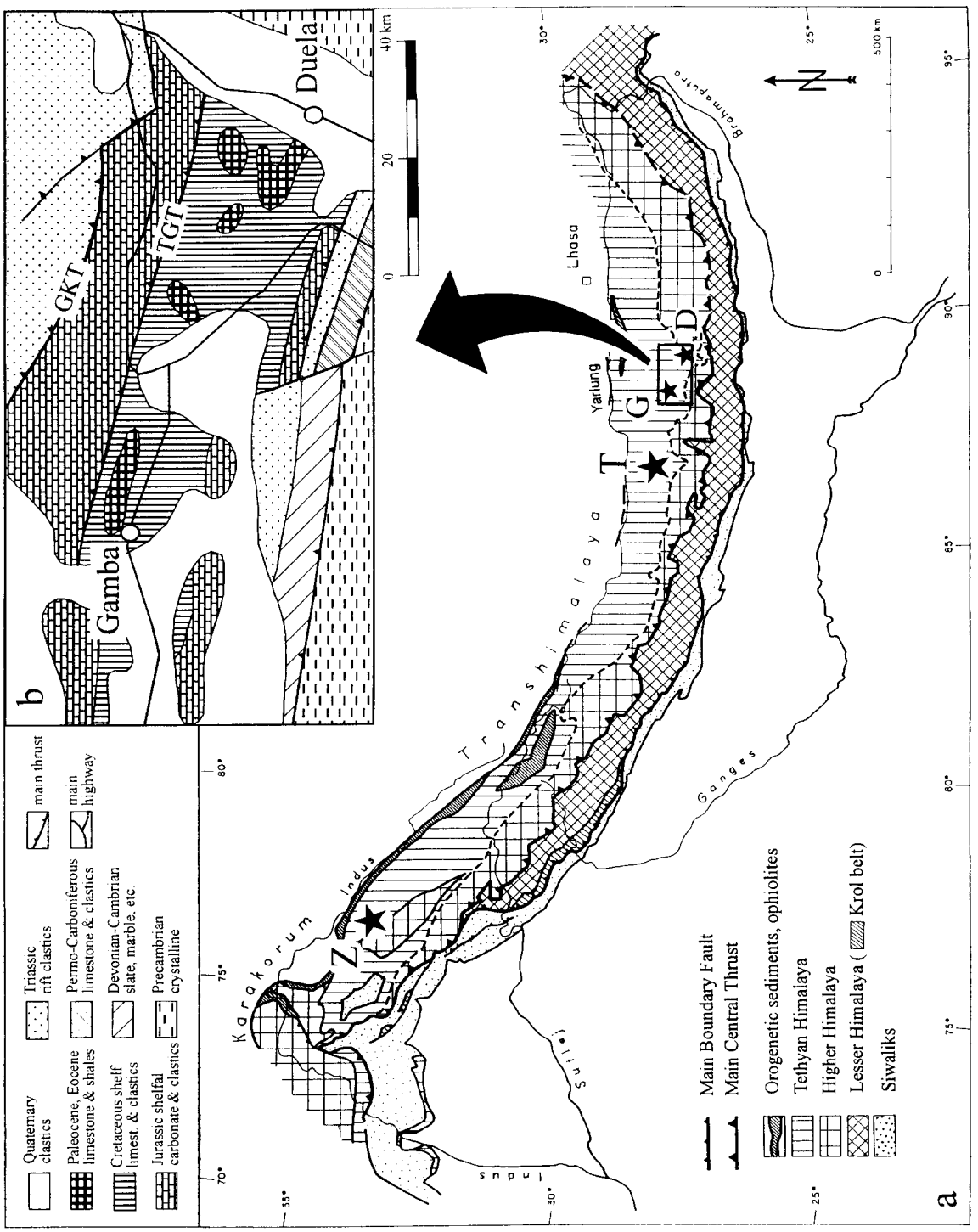
The results have been compared with expected palaeodirections according to available data for the Indian subcontinent based on its present-day geometry. The late Maastrichtian sites (fourteen sites, 156 specimens) and the middle–Late Palaeocene sites (fourteen sites, 113 specimens) yield palaeolatitudes of 5.7°S and 4.0°N for the sampling localities in southern Tibet, respectively. These palaeolatitudes are much further north than expected and imply that crustal shortening between the Tethyan Himalaya and the Indian Shield since the time of India–Eurasia collision has occurred on the order of 1500 km at the longitude of Gamba/Duela. Comparison of our observed palaeolatitudes with palaeolatitudes for the Lhasa block during Cretaceous and Palaeocene times indicate that initial contact of the northern Indian margin with the Lhasa block was already established at 65–60 Ma with suturing completed at about 55–50 Ma.

1. Introduction

The India–Eurasia convergence is well documented by apparent polar wander paths (APWP) for both continents (Klootwijk, 1984; Besse and Courtillot, 1988, 1991). The separation of the Indian Plate from Gondwana started in mid- to Late Cretaceous times by opening of the Indian ocean at a moderate southern latitude. During the Late Cretaceous to Eocene India drifted northward over about 4000 km with drift rates up to 15–20 cm/yr. The drift is well documented from analyses of magnetic anomalies

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b
Gamba
Duela
GKT
TGT
0
20
40 km
Quaternary clastics
Paleocene, Eocene limestone & shales
Cretaceous shelf limest & clastics
Jurassic shelfal carbonate & clastics
Triassic rift clastics
Permo-Carboniferous limestone & clastics
Devonian-Cambrian slate, marble, etc.
Precambrian crystalline
main thrust
main highway
Karakorum
Indus
Transhimalaya
Lhasa
Yarlung
Brahmaputra
Ganges
Indus
Sutlej
Main Boundary Fault
Main Central Thrust
Orogenetic sediments, ophiolites
Tethyan Himalaya
Higher Himalaya
Lesser Himalaya (Krol belt)
Siwaliks
0
500 km
N
a

from the Indian ocean (McKenzie and Sclater, 1971; Klootwijk and Peirce, 1979; Patriat and Achache, 1984; Klootwijk et al., 1985; Royer and Sandwell, 1989). A significant drop in the drift rate from nearly 20 to less than 5 cm/yr at 55–50 Ma is mostly interpreted as the onset of the collision of the Indian Plate with the Eurasian Plate (e.g., Patriat and Achache, 1984; Besse and Courtillot, 1988). The initial collision occurred in the western part with progressive closure of the Neo-Tethyan ocean from west to east (Klootwijk et al., 1985). Further convergence led to straightforward and rotational underthrusting of the Indian Plate beneath the southern Eurasian margin since the Miocene. During the past years an earlier age for the collision has been proposed. Based on a review of primary and secondary palaeoremanences from the Himalaya and Hindukush and results from the Ninetyeast Ridge, Klootwijk et al. (1991, 1992) concluded that initial contact between India and Eurasia was already established at the Cretaceous/Tertiary time boundary. They interpret the drop in drift rate at 55–50 Ma as completion of closure of the Neo-Tethys. Occurrence of high-pressure rocks of eclogite facies in the Upper Kaghan Valley (northwestern Himalaya) with a peak metamorphic age of 49 ± 6 Ma (Tonarini et al., 1993; Spencer, 1993) supports the hypothesis of an initial collision as early as 65 Ma in the western part. A vertebrate fauna with distinct Laurasian affinities found in intertrappean sediments, interbedded in the Deccan traps (Cretaceous/Tertiary boundary) give evidence for such an early contact (Prasad and Sahni, 1988; Jaeger et al., 1989).

Models for the mechanism of the India–Eurasia collision and for crustal shortening within the collision zone have been proposed by many authors (e.g., Powell and Conaghan, 1973; Le Fort, 1975; Seeber et al., 1981; Tapponnier et al., 1982; Allegre et al., 1984; Patriat and Achache, 1984; Klootwijk et al., 1985, 1991; Powell, 1986; Tapponnier et al., 1986; Dewey et al., 1988; Chen et al., 1993b). Crustal shortening through imbricate stacking within the Himalayan crust, underplating of the Indian Plate beneath Eurasia, thickening of the Eurasian crust through plastic deformation and folding, and eastward extrusion of continental blocks within southern Eurasia are discussed. For quantitative reconstructions and for dating of the initial contact, knowledge about the extent of the former northern margin of India, the so-called Greater India (Veevers et al., 1975), is essential. The Tethyan Himalaya, as the former northern margin of the Indian Plate plays a key role in reconstructions of Greater India, palaeomagnetic data from this zone, however, are quite rare. Results have been published by Klootwijk and Bingham (1980), Klootwijk et al. (1983), Zhu Zhiwen and Teng Jiwen (1984), Besse et al. (1984) and Appel et al. (1991, 1995), but only a single study (Tingri, 29°N, 87°E, Besse et al., 1984) details a palaeoremanence with an acquisition age within the particularly important period of rapid northward drift of India. Apart from Tingri, Late Cretaceous–Eocene marine Tethyan sediments are found only in Zanskar (34°N, 76.5°E) and in the area around Gamba and Duela (28°N, 89°E) (see Fig. 1a). Intensive palaeomagnetic investigations in Zanskar (Appel et al., 1995) yielded a well established secondary palaeoremanence, but no primary component.

Results from the Tethyan Himalaya of Gamba and Duela are presented here for middle Albian–Late Palaeocene rocks and provide information on palaeolatitudes and rotations of the northernmost part of Greater India prior to and during the early stages of the collision of India with Eurasia.

2. Geology and sampling

The principal tectonic units of the Himalaya are shown in Fig. 1a. The suture zone (IYS: Indus Yarlung suture) between India and Eurasia is marked by ophiolites along the rivers Indus and Yarlung. During further convergence the active zone of subduction propagated southward leading to continental underthrusting along the Main Central Thrust (MCT) in Oligocene–Miocene times, subsequently along the

Fig. 1. (a) Schematic map of the Himalaya (simplified after Gansser, 1964). The localities with Late Cretaceous–Eocene marine Tethyan sediments are marked by stars. Z = Zanskar; T = Tingri; G = Gamba; D = Duela. (b) Tectonostratigraphic map of the investigation area (after Liu, 1992). *GKT* = Gyirong–Kangmar thrust; *TGT* = Tingri–northern Gamba thrust.

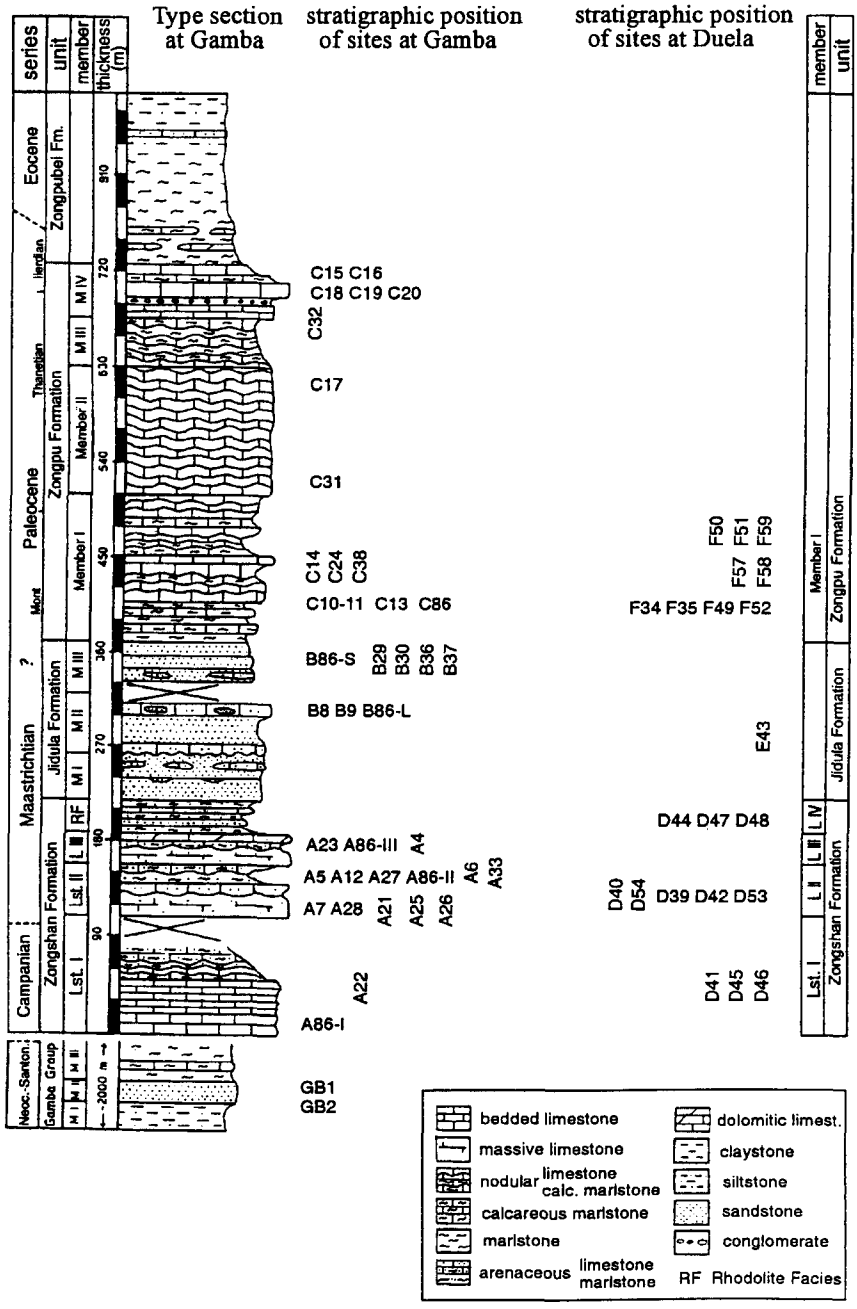


Fig. 2. Lithostratigraphic type section of the Cretaceous–early Tertiary stratigraphic succession of the Tethyan Himalaya in the Gamba area (after Willems and Zhang, 1993), and schematic stratigraphy of the corresponding succession at Duela (Pflästerer et al., 1993). The stratigraphic position of sites are indicated horizontally (sites from which significant results could be obtained) or vertically (no results obtained).

Main Boundary Fault (MBF) (e.g., Le Fort, 1975), and finally along the presently active Main Frontal Thrust (Gansser, 1991). The Tethyan Himalaya as the former leading edge of Greater India is now situated directly south of the IYS. The sedimentary sequences of the Tethyan zone were deposited as shelf sediments on the northern margin of the Indian continent from Precambrian to Eocene times.

Our sampling areas near Gamba (28.3°N, 88.5°E) and Duela (28.0°N, 89.3°E) are shown in Fig. 1b. In the Gamba area a continuous sedimentary succession from late Aptian to Early Eocene is well preserved (Gansser, 1964; Wen, 1987a,b; Willems and Zhang, 1993). A detailed lithological and biostratigraphic description of this succession is given by Willems and Zhang (1993). No signs of metamorphism are found within these sediments and occurrence of anatas (TiO_2 modification) in the Jidula sandstones of Gamba and Duela proves that heating did not exceed 260°C (J. Schaller, pers. commun., 1994).

Our work focused on the Albian–Palaeocene successions, deposited during the rapid northward drift of India prior to and during the onset of collision. The mid-Cretaceous–Palaeocene succession of the type section at Gamba is shown in Fig. 2. It can be divided into four lithostratigraphic units: the Gamba Group (Neocomian–Santonian), the Zongshan Formation (Campanian–Maastrichtian), the Jidula Formation (latest Maastrichtian–Early Palaeocene) and the Zongpu Formation (middle Palaeocene–latest Palaeocene). The Gamba Group can be divided into three members and consists of gray shales with

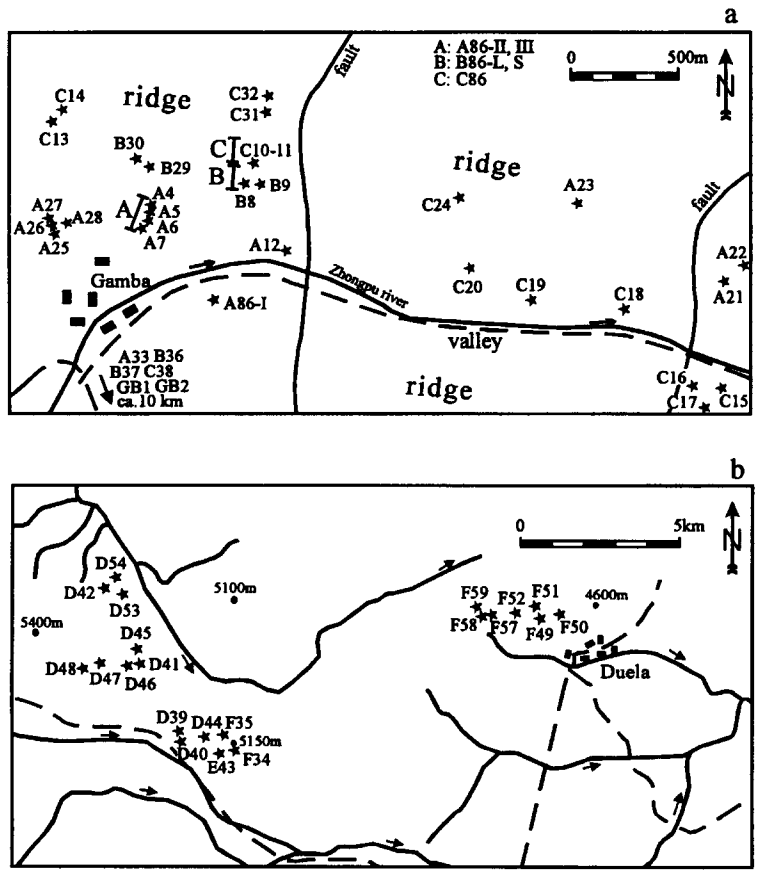


Fig. 3. Local maps of the sampling areas together with the site locations. The sites are marked by stars, or bars for the sampled profiles, with the corresponding site name. Roads are marked by stipple lines, rivers by heavy lines and arrows. (a) Local map of Gamba (28.3°N, 88.5°E). (b) Local map of Duela (28.0°N, 89.2°E).

interbedded marls, sandstones and limestones. The overlying Zongshan Formation consists of three scarp-forming limestone units with interbedded calcareous marls. It is very rich in fossils, allowing excellent biostratigraphic classification. The Jidula Formation is built up mainly by sandstones, with a dark limestone layer intercalated in Member II. The Cretaceous/Tertiary boundary occurs within this formation. Its exact position, however, is unclear due to the lack of biostratigraphically significant fossils in the sandstones. The Zongpu Formation consists mainly of massive limestone beds, nodular limestones and calcareous marls. Planktonic foraminifera of the Angulata Zone indicate a middle Palaeocene age for the lower part of the Zongpu Formation. The occurrence of Alveolina, Orbitolites and small Nummulites in Member IV, the top of the Zongpu Formation, give a classification into the Ilerdian. The Palaeocene/Eocene boundary lies within the overlying Zongpubei Formation, which consists of greenish marls and red claystones and siltstones. The lithostratigraphy of the Late Cretaceous–Palaeocene succession in Duela is generally comparable to that of Gamba (Pflästerer et al., 1993). However, the Jidula limestone layer is not exposed at Duela.

Sampling was carried out in the summers of 1986 and 1992 as part of Chinese–German joint projects for study of the Cretaceous/Tertiary stratigraphy and evolution of the Tethyan Himalaya in southern Tibet. A portable rockdrill was used for sampling, and the cores were oriented with a magnetic compass. The generally weak intensity of magnetization in the rocks did not require the use of a solar compass. Usually ten cores of 2.5 cm diameter were taken from each site. A total of more than 700 oriented samples from 61 sites were drilled from the above mentioned four stratigraphic units of mid-Cretaceous–latest Palaeocene age at Gamba and Duela. No samples could be taken from the highly weathered Eocene Zongpubei Formation. The stratigraphic positions of the sites are shown in Fig. 2. Local maps of the sampling areas together with the exact site locations are shown in Fig. 3a and b.

3. Sample treatment

The core samples were cut into specimens with a length of 2.2–2.3 cm. For analyses of the ferrimagnetic minerals one specimen per site was used for isothermal remanence acquisition (IRM) studies in a DC field up to 1.5 T with subsequent stepwise thermal demagnetization of the saturation IRM (SIRM). Detailed stepwise alternating field (AF) and

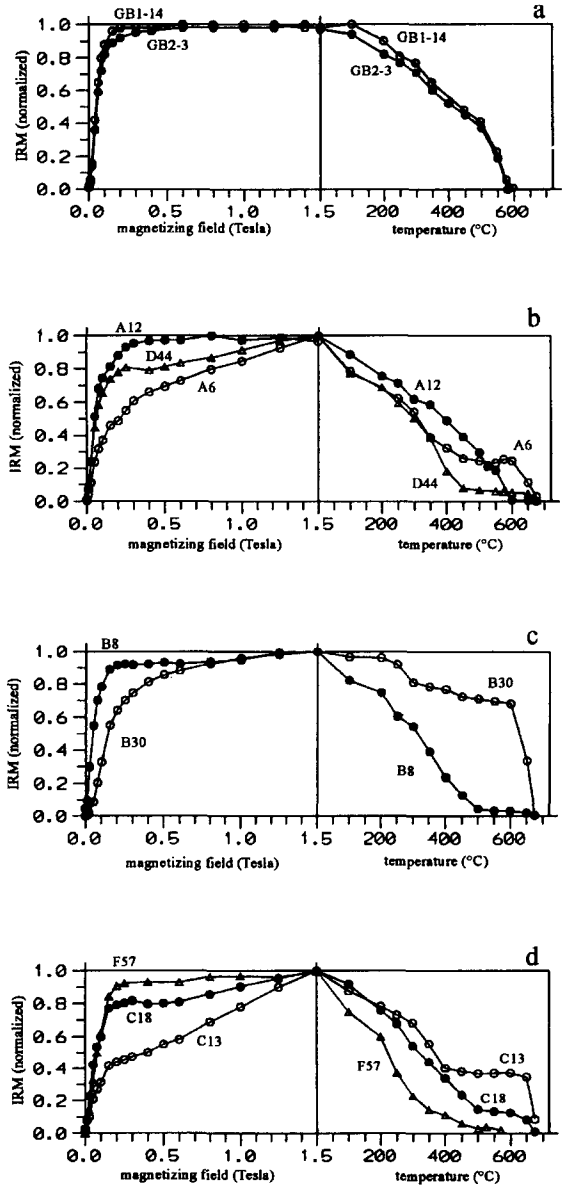


Fig. 4. Acquisition and thermal demagnetization curves of isothermal remanent magnetization for representative specimens of the investigated stratigraphic units. (a) Gamba Group. (b) Zongshan Formation. (c) Jidula Formation. (d) Zongpu Formation.

thermal demagnetization (TH) was carried out on pilot specimens from each site. Susceptibility was monitored after each heating step to detect possible mineralogical alterations. Depending on the results of the pilot studies and the IRM curves, the remaining specimens of all sites were demagnetized according to the best suitable procedure; that is either thermally (MTDM1 and Schonstedt furnaces), or in alternating fields (using 2G and Highmoor AF demagnetizers), or a combination of both methods. Remanence directions were measured with cryogenic magnetometers (Rémanomètre à Squid RS 01, LETI, noise level about 0.02 mA/m; 2G 755 R, noise level < 0.01 mA/m). Orthogonal vector plots and equal area projections were used to examine the behaviour of the specimens upon demagnetization. Magnetic directions were mainly ($> 70\%$) determined from straight segments of the demagnetization paths through principal component analysis (PCA, modified version after Kirschvink, 1980), and in part ($< 30\%$) from stable end directions. Calculations of mean values were done applying Fisherian statistics (Fisher, 1953).

4. Palaeomagnetic results

The results from both sampling localities, Gamba and Duela, are discussed together for each lithostratigraphic unit, because their rock magnetic properties and demagnetization behaviour are very similar.

4.1. The Gamba Group (Member II)

Magnetite is the only ferrimagnetic mineral indicated by the low coercivity forces and the unblocking temperatures of about 570°C (Fig. 4a). Therefore, the bulk of the specimens were AF demagnetized. Typically a two-component behaviour could be observed (Fig. 5a and b). A soft, randomly oriented magnetic remanence was completely removed at peak fields of 5–20 mT. Statistical data for the remaining characteristic remanence (ChRM) directions are listed in Table 1. Both sampled sites yielded significant results. The single specimen directions for the ChRM and the in situ directions for the soft component are shown in Fig. 6. The site mean values are plotted in Fig. 13.

4.2. The Zongshan Formation

Nearly all sites show IRM acquisition curves (Fig. 4b) with a lower coercivity component (exception sites A26, D40) that is in many cases coexisting with a high coercivity component. Thermal demagnetization of the SIRM (Fig. 4b) and NRM (Fig. 7e) reveals unblocking temperatures for the two components up to 450–580°C and 675°C, respectively,

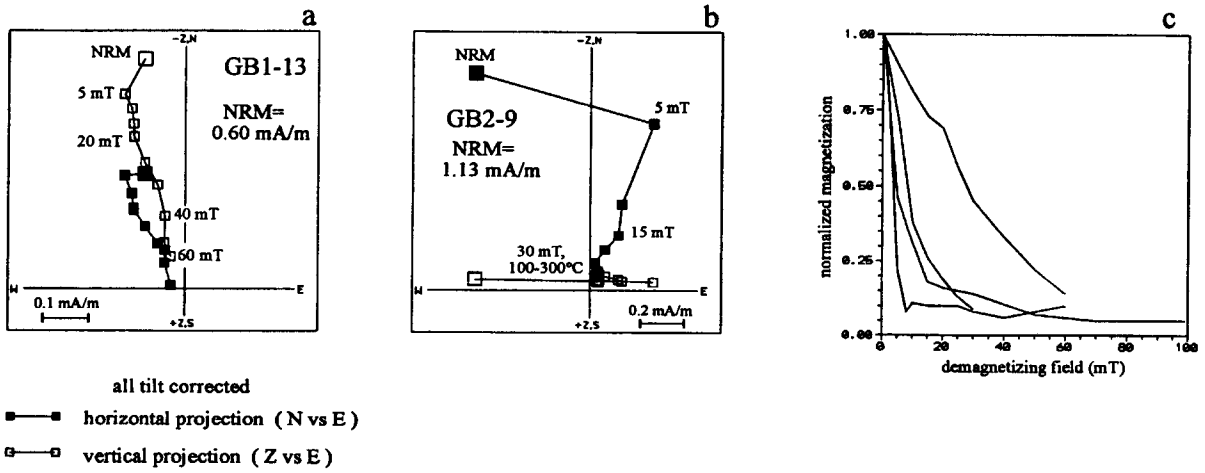


Fig. 5. Demagnetization results of representative specimens from the Gamba Group. (a, b) Orthogonal vector projections. (c) Decay of remanence intensities during AF demagnetization.

Table 1
Statistical parameters for the Gamba Group (Member II) from Gamba

Site name	Lithology	Geologic age	Numerical age (Ma)	N	Spec. n/r	Bedding	In situ		Tilt corrected		α_{95}	k	Palaeo-latitude	North pole position			
							Dec.	Inc.	Dec.	Inc.				Long.	Lat.	dp	dm
GB1	graywacke	mid.–late Albian	98–107	13	9/0	356,60	351.2	20.9	350.2	−38.9	15.6	11.8	22.0°S	280.2	38.8	11.1	18.6
GB2	graywacke	mid.–late Albian	98–107	15	14/0	3,65	353.5	24.6	351.8	−39.6	9.7	17.9	22.5°S	278.2	38.6	7.0	11.6

Numerical ages are based on the time table of Ogg (1995); N = number of specimens measured; spec. (n/r) = number of specimens used in the statistics with normal (n) or reverse (r) polarity; bedding = azimuth and dip of the dip arrow (azimuth 90° means east); Dec./Inc. = palaeodeclination/palaeoinclination; α_{95} = confidence angle; k = precision parameter; palaeolatitudes = calculated for Gamba (28.3°N, 88.5°E); long./lat. = longitude/latitude; dp / dm = confidence limits of poles calculated from tilt corrected site mean directions.

Table 2
Statistical parameters for the Zongshan Formation from Gamba

Site name	Lithology	Geologic age	Numerical age (Ma)	N	Spec. n/r	Bedding	In situ		Tilt corrected		α_{95}	k	Palaeo-latitude	North pole position			
							Dec.	Inc.	Dec.	Inc.				Long.	Lat.	dp	dm
A86-I	limestone	early Campan.	80–83	12	8 / 0	31,29	356.8	−19.7	345.7	−42.3	14.4	15.8	24.5°S	184.5	35.5	10.9	17.7
A22	limestone	Campanian	71–83	10		17,35			–	–							
A4	limestone	late Maastricht.	65–69	11		42,16			–	–							
A5	limestone	late Maastricht.	66–69	17	11 / 0	24,25	354.2	−0.7	351.4	−22.8	13.4	12.7	11.9°S	281.4	49.0	7.5	14.2
A6	limestone	late Maastricht.	66–69	14		36,23			–	–							
A7	limestone	late Maastricht.	66–69	14	12 / 0	36,23	15.7	−7.9	12.8	−29.3	11.1	16.4	11.1°S	251.2	44.3	6.8	12.3
A12	limestone	late Maastricht.	66–69	17	10 / 0	31,26	25.4	7.3	25.2	−18.6	13.8	13.2	9.6°S	232.1	45.0	7.5	14.4
A21	limestone	late Maastricht.	66–69	11		61,22			–	–							
A23	limestone	late Maastricht.	65–69	11	10 / 0	36,146[a]	194.2	−16.4	14.3	−15.4	9.7	25.8	7.9°S	245.4	51.3	5.1	10.0
A25	limestone	late Maastricht.	66–69	11		65,22			–	–							
A26	limestone	late Maastricht.	66–69	17		65,22			–	–							
A27	limestone	late Maastricht.	66–69	14	8 / 0	71,19	10.7	−13.7	5.0	−22.4	10.2	30.3	11.6°S	260.9	49.8	5.7	10.8
A28	limestone	late Maastricht.	66–69	10	6 / 0	62,25	7.2	−11.0	0.4	−24.3	21.8	10.4	12.7°S	267.9	49.0	12.5	23.3
A33	limestone	late Maastricht.	66–69	8		339,22			–	–							
A86-II	limestone	late Maastricht.	66–69	51	31 / 0	39,20	5.5	0.9	3.6	−15.2	7.9	11.6	7.7°S	262.5	53.8	4.2	8.1
A86-III	limestone	late Maastricht.	65–69	40	18 / 4	49,20	357.9	20.9	1.8	7.5	11.1	8.7	3.8°N	264.2	65.4	5.6	11.2

Explanations and abbreviations see Table 1; – = no significant result.[a] Overturned.

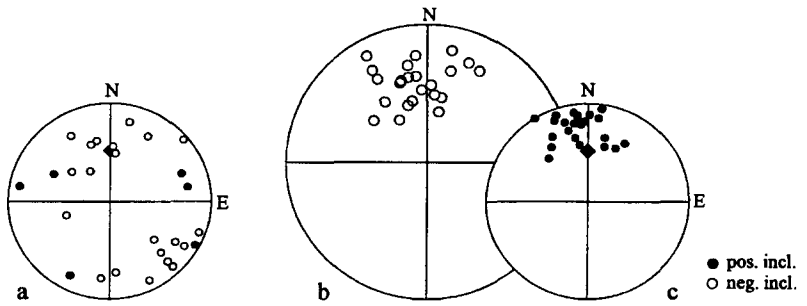


Fig. 6. Equal area projections of individual specimen directions from the Gamba Group. (a) In situ soft components. (b) Tilt corrected ChRM directions. (c) In situ ChRM directions. The present dipole field direction is shown as a rhomb in (a) and (c).

indicating magnetite and hematite as ferrimagnetic minerals. In some sites goethite is also identified by a significant decay of the SIRM at about 100°C. Analyses of directional data show a two-component behaviour. During AF demagnetization (Fig. 7a) a soft component with an in situ direction parallel to the local ambient field is removed up to about 15 mT. A second, harder component is demagnetized between 20 and 100 mT, with the demagnetization path pointing to the origin. A similar two-component

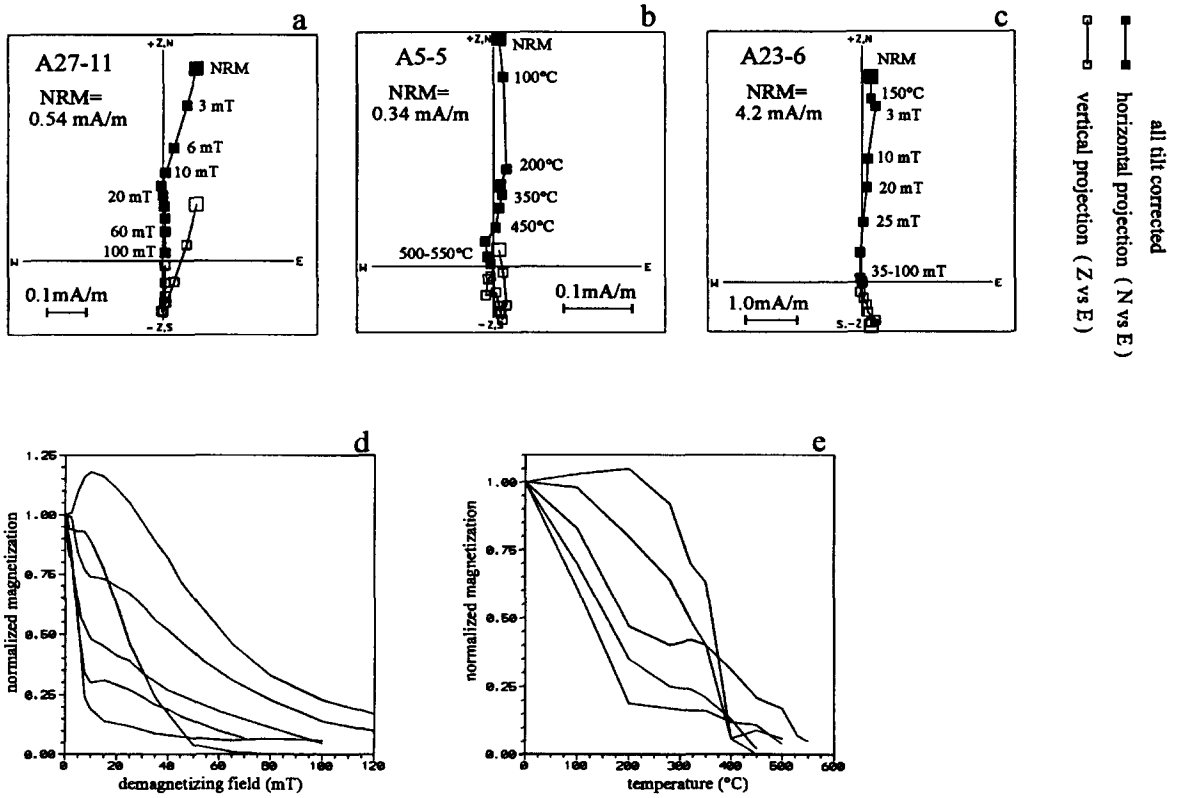


Fig. 7. Demagnetization results of representative specimens from the Zongshan Formation. (a–c) Orthogonal vector projections. (d–e) Remanence intensities.

Table 3
Statistical parameters for the Zongshan Formation from Duela

Site name	Lithology	Geologic age	Numerical age (Ma)	N	Spec. n/r	Bedding	In situ		Tilt corrected		α_{95}	k	Palaeo-latitude	North pole position			
							Dec.	Inc.	Dec.	Inc.				Long.	Lat.	dp	dm
D41	limestone	Campanian	71–84	9		345,5			–	–							
D45	limestone	Campanian	71–84	11		142,7			–	–							
D46	limestone	Campanian	71–84	10		16,15			–	–							
D39	limestone	Maastrichtian	65–71	11	0/8	352,18	176.9	−4.2	177.0	13.9	25.7	5.6	7.1°S	274.4	54.8	13.4	26.3
D40	limestone	Maastrichtian	65–71	10		346,20			–	–							
D42	limestone	Maastrichtian	65–71	11	0/10	217,24	195.7	8.2	195.2	−14.6	4.2	130.4	7.4°N	231.4	64.9	2.2	4.3
D53	limestone	Maastrichtian	65–71	11	0/6	215,24	188.3	22.5	190.5	1.2	17.2	16.1	0.6°S	248.0	59.7	8.6	17.2
D54	limestone	Maastrichtian	65–71	9		76,27			–	–							
D44	limestone	late Maastricht.	65–69	9	0/8	45,12	167.9	−13.9	169.7	−7.3	5.7	95.1	3.7°N	293.0	63.8	2.9	5.7
D47	limestone	late Maastricht.	65–69	9	0/7	301,27	169.1	−3.8	171.0	14.2	14.9	17.4	7.2°S	284.4	53.7	7.8	15.2
D48	limestone	late Maastricht.	65–69	8	0/7	47,30	176.1	−11.1	176.6	8.4	15.0	17.2	4.2°S	275.5	57.6	7.6	15.1

Explanations and abbreviations see Table 1; – = no significant result.

Table 4
Statistical parameters for the Jidula Formation

Site name	Lithology	Geologic age	Numerical age (Ma)	N	Spec. n/r	Bedding	In situ		Tilt corrected		α_{95}	k	Palaeo-latitude	North pole position			
							Dec.	Inc.	Dec.	Inc.				Long.	Lat.	dp	dm
B8	limestone	C/T-boundary	64–66	9	0/7	55,18	168.4	−18.6	172.8	−10.6	9.1	45.1	5.3°N	286.4	66.1	4.7	9.2
B9	limestone	C/T-boundary	64–66	9	0/9	55,18	169.0	−23.7	174.9	−15.2	4.9	113.5	7.7°N	282.6	68.9	2.6	5.0
B86-L	limestone	C/T-boundary	64–66	4	0/3	55,18	178.4	−24.1	183.6	−12.9	16.8	54.6	6.5°N	258.9	68.0	8.7	17.1
B29	sandstone	C/T-boundary	63–65	10		77,25			–	–							
B30	sandstone	C/T-boundary	63–65	10		60,30			–	–							
B36	sandstone	C/T-boundary	63–65	11		337,22			–	–							
B37	sandstone	C/T-boundary	63–65	11		357,23			–	–							
B86-S	sandstone	C/T-boundary	63–65	44	31/6	32,28	354.6	42.6	3.2	18.6	8.0	9.6	9.6°N	258.8	71.0	4.3	8.3
E43	sandstone	C/T-boundary	63–65	12		11,20			–	–							

Explanations and abbreviations see Table 1; – = no significant result.

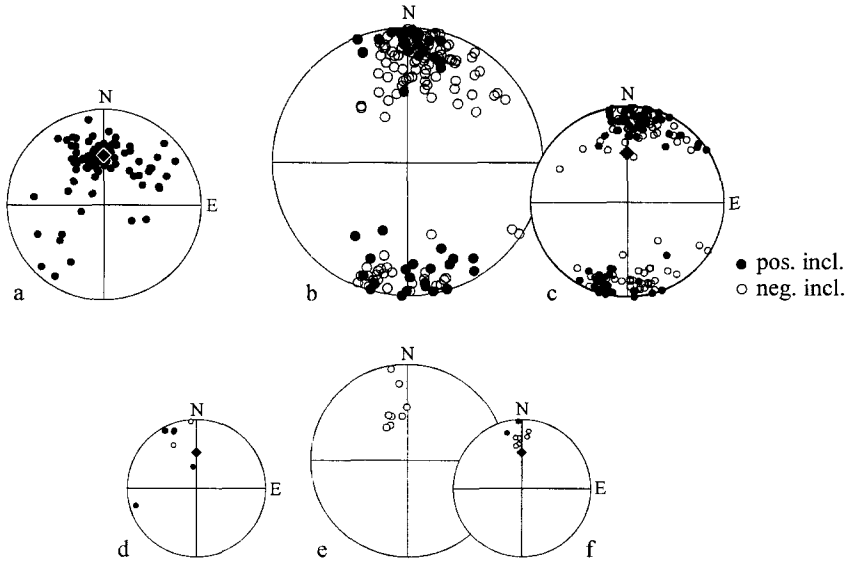


Fig. 8. Equal area projection of individual specimen directions from the Zongshan Formation. (a) In situ soft components. (b) Tilt corrected ChRM directions. (c) In situ ChRM directions from members II, III, IV. (d) In situ soft components. (e) Tilt corrected ChRM directions. (f) In situ ChRM directions from Member I. The present dipole field direction is shown as a rhomb in (a), (c), (d) and (f).

behaviour is observed during thermal treatment (Fig. 7b), but the demagnetization path is less smooth. The soft component is removed up to about 350°C, the hard component up to about 530°C. Generally a major increase in susceptibility is observed above 450°C. The bulk of the specimens from the Zongshan Formation were therefore first heated up to 150°C in order to destroy possible goethite remanences and subsequently AF demagnetized in fields up to 100 mT. The hard AF component was taken as the ChRM. In some specimens only a single component behaviour was observed (Fig. 7c). In the case of low median destructive fields (MDF) < 10 mT, the specimens were rejected from further calculations. Single components with a high MDF > 15 mT were accepted as representing the ChRM (Fig. 7c, the magnetization intensity remaining after the 150°C step is taken as the initial value for calculating the MDF).

The demagnetization behaviour together with IRM curves prove magnetite as the carrier of the ChRM. The high stability of this harder magnetite component upon AF demagnetization (Fig. 7a–d) suggests a single-domain (SD) or pseudo-single-domain (PSD) status of this magnetite fraction. Hematite does not contribute significantly to the total remanence. In cases of weak NRM intensities a residual hematite component is partially observed after the maximum step of demagnetization. However, ChRM directions at 20 mT (dominated by magnetite) do not significantly differ from the stable residual directions at 100 mT (mainly carried by hematite).

Statistical data for ChRM directions are listed in Table 2 for the Gamba sites and in Table 3 for the Duela sites. Fifteen of the 27 treated sites yielded significant results. The in situ directions of the soft component and the single specimen directions of the ChRM are plotted in Fig. 8. The site mean values are plotted in Fig. 13. Rejected sites either show a present field direction only, or a high scattering of remanence directions. The implications of the opposing mainly normal and exclusively reverse polarities in the Gamba and Duela sites, respectively, will be discussed below.

4.3. The Jidula Formation

The lithologic differences of Jidula members I and III (sandstones) and Jidula Member II (limestones) result in a different magnetic mineralogy.

Hematite is predominant in the sandstones with additional goethite and magnetite in some of the

specimens [identified mainly by unblocking temperatures above 600°C, at about 100°C and at about 300–500°C, respectively (Fig. 4c, specimen B30; Fig. 9a–c)]. After removal of goethite remanences through heating above 100°C, remanence directions remained stable up to about 650°C (Fig. 9a and b). ChRM directions, carried by hematite (and probably also by magnetite in some of the specimens) were determined from demagnetization data obtained after the 200°C heating step.

In the limestones magnetite is dominant with a minor hematite component, identified by low coercivity combined with unblocking temperatures < around 500°C and high coercivity combined with unblocking temperatures > 600°C, respectively (Fig. 4c, specimen B8). AF demagnetization usually revealed two components (Fig. 9d–f). After removal of a soft component with scattered directions a harder component is demagnetized between 20 and 100 mT. This hard component is taken as the ChRM, similar to the Zongshan Formation. The ChRM could be determined from most of the specimens. As it was the case for the Zongshan limestones hematite does not contribute significantly to the total remanence.

Statistical data for the ChRM directions are listed in Table 4. Individual specimen directions for the ChRM component and also for the soft component of the limestones are shown in Fig. 10. The site mean values are plotted in Fig. 13. Reliable results are obtained from four sites (three limestone, one sandstone) from the Gamba area. With the exception of one site, the sandstones show highly scattered

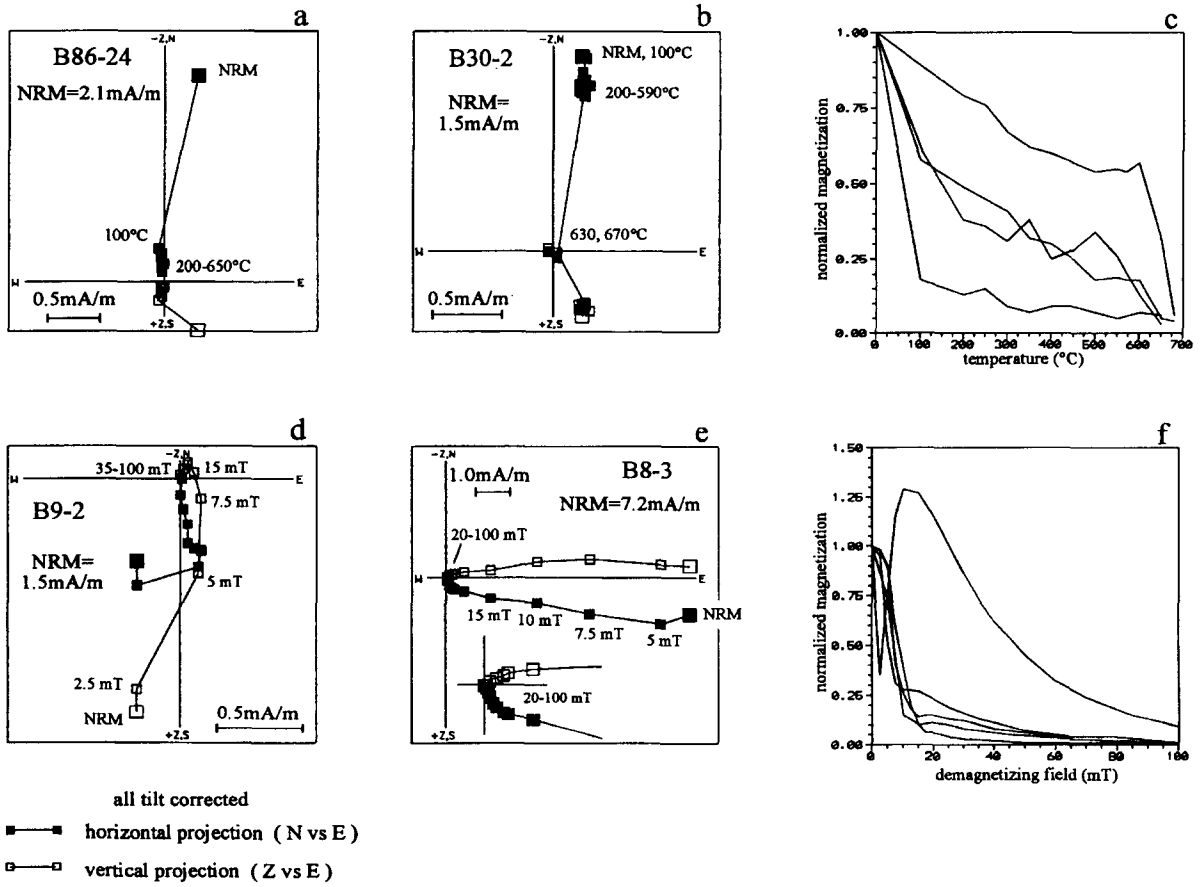


Fig. 9. Demagnetization results of representative specimens (sandstones) from the Jidula Formation Member III (a–c) and representative specimens (limestones) from the Jidula Formation Member II (d–f). (a, b ,d, e) Orthogonal vector projections. (c, f) Remanence intensities.

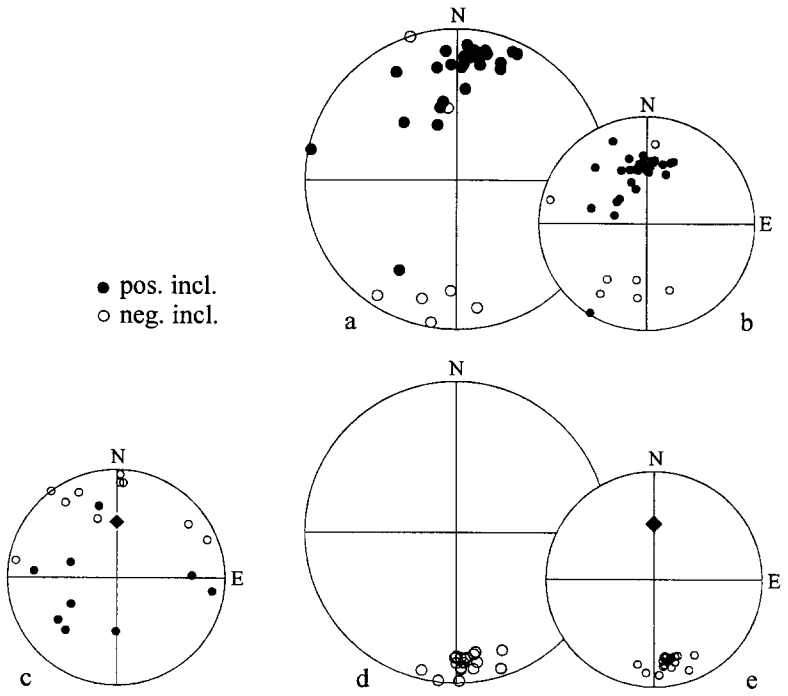


Fig. 10. Equal area projection of individual specimen directions from the Jidula Formation. (a) Tilt corrected ChRM directions. (b) In situ ChRM directions from Member III (sandstones). (c) In situ soft components. (d) Tilt corrected ChRM directions. (e) In situ ChRM directions from Member II (limestones). The present dipole field direction is shown as a rhomb in (b), (c) and (e).

directions. In contrast to the sandstones, the limestones of the Jidula Member II provide excellent results. Unfortunately, this limestone layer is observed only in the Gamba region over a small lateral extension of about 500 m.

4.4. The Zongpu Formation

IRM curves (Fig. 4d) and demagnetization behaviour (Fig. 11) are similar to the Zongshan Formation. Low and high coercivity forces and unblocking temperatures of 400–500°C and 675°C indicate magnetite and hematite as ferrimagnetic minerals. Goethite also occurs in several sites.

Similar to the Zongshan Formation the susceptibility increased during thermal demagnetization above about 450°C. The bulk of the specimens was demagnetized in alternating fields up to 100 mT after a single heating step at 120 or 150°C. A soft component, parallel to the local ambient field, was then removed up to 15 mT. A very well determined harder component was demagnetized between 20 and 100 mT, with a demagnetization path pointing to the origin. This component is interpreted as the ChRM direction (Fig. 11a and b). The high stability of the hard component upon AF demagnetization (Fig. 11c) indicates a SD or PSD domain status of this magnetite fraction. Similar to the Zongshan Formation magnetite is the carrier of the ChRM and remanences carried by hematite seem to be negligible.

Statistical data for the ChRM directions are listed in Table 5 (Gamba) and Table 6 (Duela). Fourteen of the 23 treated sites yielded reliable results. Detailed stratigraphic control allows us to divide the results from Gamba in a "lower" (middle Palaeocene) and "upper" (Late Palaeocene) Zongpu Group with four and six sites, respectively (see also Fig. 2). Individual specimen directions for the ChRM and the soft component are plotted in Fig. 12. The site mean directions are plotted in Fig. 13. Sites were rejected if they showed either a present field direction only (mainly due to low NRM intensities), or a high scatter of remanence directions.

Table 5
Statistical parameters for the Zongpu Formation from Gamba

Site name	Lithology	Geologic age	Numerical age (Ma)	N	Spec. n/r	Bedding	In situ		Tilt corrected		α_{95}	k	Palaeo-latitude	North pole position			
							Dec.	Inc.	Dec.	Inc.				Long.	Lat.	dp	dm
Lower Zongpu																	
C10-11	limestone	mid. Palaeocene	58–63	17	2/5	25,38	170.7	−36.6	178.4	−3.9	14.6	18.0	2.0°N	272.1	63.6	7.3	14.6
C13	limestone	mid. Palaeocene	58–63	11	0/9	12,35	166.8	−32.1	171.0	0.3	9.1	32.8	0.2°S	286.9	60.3	4.6	9.1
C14	limestone	mid. Palaeocene	58–63	11		12,35			–	–							
C24	limestone	mid. Palaeocene	58–63	8		27,20			–	–							
C31	limestone	mid. Palaeocene	58–63	9	0/5	182,21	183.9	19.2	183.8	−1.7	21.4	13.7	0.9°N	260.3	62.3	10.7	21.4
C38	limestone	mid. Palaeocene	58–63	11		6,22			–	–							
C86	limestone	mid. Palaeocene	59–61	24	0/21	52,25	161.2	−29.4	171.6	−18.7	5.2	38.2	9.6°N	293.6	69.8	2.8	5.4
Upper Zongpu																	
C15	limestone	Late Palaeocene	55–60	12	0/7	42,25	182.5	−31.8	188.4	−12.0	9.1	44.5	6.1°N	247.2	66.4	4.7	9.3
C16	limestone	Late Palaeocene	55–60	11	0/95	5,31	186.1	−28.6	193.3	−6.8	10.4	25.7	3.4°N	239.1	62.1	5.3	10.5
C17	limestone	Late Palaeocene	55–60	10	0/8	37,28	174.4	−41.7	184.8	−19.0	4.8	135.1	9.8°N	253.9	70.9	2.6	5.0
C18	limestone	Late Palaeocene	55–60	11	0/10	25,29	167.4	−27.3	172.2	−3.4	17.1	9.0	1.7°N	285.5	62.4	8.6	17.1
C19	limestone	Late Palaeocene	55–60	12	0/7	29,14	187.7	−28.4	189.6	−15.9	19.8	10.2	8.1°N	242.5	67.9	10.5	20.4
C20	limestone	Late Palaeocene	55–60	12	0/7	69,14	181.6	−22.4	186.3	−16.5	10.4	34.4	8.4°N	250.7	69.3	5.5	10.7
C32	limestone	Late Palaeocene	55–60	9		159,22			–	–							

Explanations and abbreviations see Table 1; – = no significant result.

Table 6
Statistical parameters for the Zongpu Formation from Duela

Site name	Lithology	Geologic age	Numerical age (Ma)	N	Spec. n/r	Bedding	In situ		Tilt corrected		α_{95}	k	Palaeo-latitude	North pole position				
							Dec.	Inc.	Dec.	Inc.				Long.	Lat.	dp	dm	dm
F34	limestone	mid. Palaeocene	58–63	10	2/3	1,27	154.6	−20.1	156.3	4.0	17.5	20.1	2.0°S	310.3	52.3	8.8	17.5	17.5
F35	limestone	mid. Palaeocene	58–63	9	0/5	177,10	174.1	20.2	174.2	9.9	17.0	21.1	5.0°S	279.7	56.5	8.7	17.2	17.2
F49	limestone	mid. Palaeocene	58–63	13	1/5	308,14	167.6	−18.1	165.6	−7.5	11.6	64.0	3.7°N	301.3	62.2	5.9	11.7	11.7
F50	limestone	mid. Palaeocene	58–63	9		322,11			–	–								
F51	limestone	mid. Palaeocene	58–63	9		187,13			–	–								
F52	limestone	mid. Palaeocene	58–63	12	0/7	126,18	151.5	2.9	152.2	−13.1	6.9	77.2	6.7°N	325.4	56.1	3.6	7.0	7.0
F57	limestone	mid. Palaeocene	58–63	10		345,20			–	–								
F58	limestone	mid. Palaeocene	58–63	9		348,24			–	–								
F59	limestone	mid. Palaeocene	58–63	11		358,20			–	–								

Explanations and abbreviations see Table 1; – = no significant result.

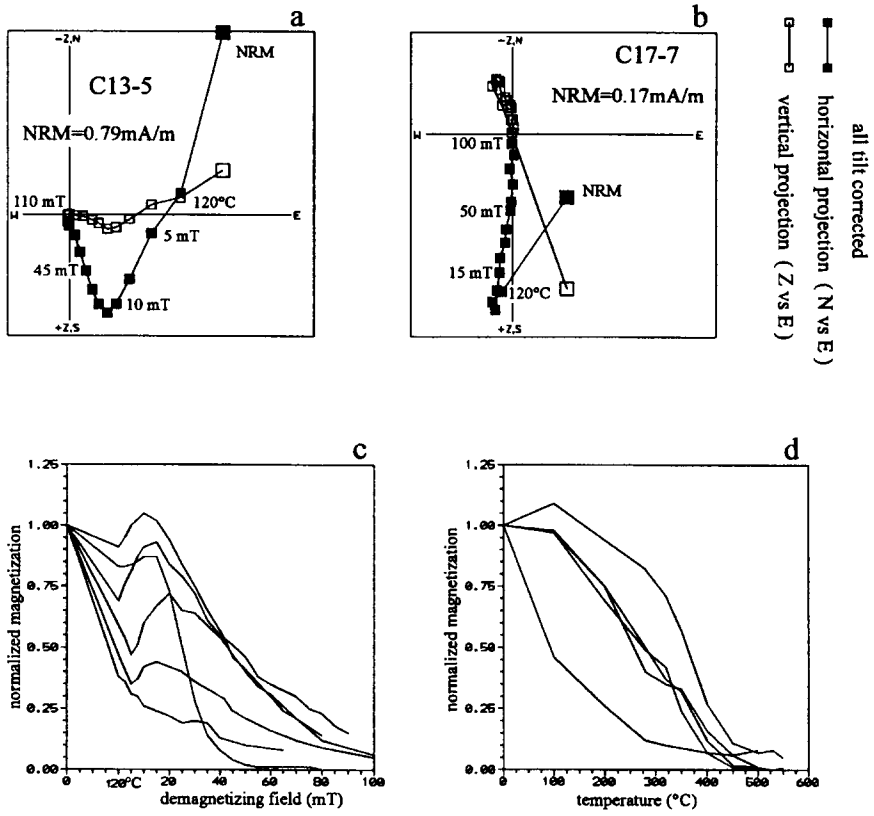


Fig. 11. Demagnetization results of representative specimens from the Zongpu Formation. (a, b) Orthogonal vector projections. (c, d) Remanence intensities.

4.5. Palaeodirections

Reliable results are obtained for 35 sites (356 specimens). Following their stratigraphic positions and sampling localities, they can be divided into eight groups (Tables 7 and 8; Fig. 13).

The ChRM certainly represents a palaeoremanence, because (1) the present magnetic field direc-

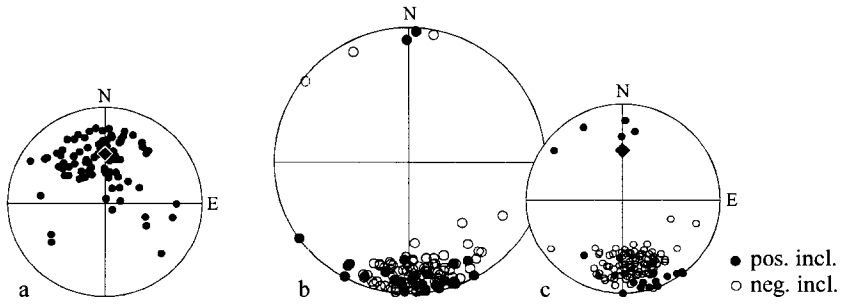


Fig. 12. Equal area projections of individual specimen directions from the Zongpu Formation. (a) In situ soft components. (b) Tilt corrected ChRM directions. (c) In situ ChRM directions. The present dipole field direction is shown as a rhomb in (a) and (c).

Table 7
Formation mean values for Gamba and Duela calculated from site mean values (site A86-I not included)

Stratigraphic unit	Lithology	Geologic age	Numerical age (Ma)	N	In situ		Tilt corrected		α_{95}	k	Palaeo-latitude	North pole position			
					Dec.	Inc.	Dec.	Inc.				Long.	Lat.	*dp*	*dm*
Gamba Group (Gamba)	graywacke	mid.–l. Albian	98–107	2	352.3	22.8	351.0	−39.3	–	–	22.3°S	279.2	38.7	–	–
Zongshan Fm. (Duela)	limestone	Maastrichtian	65–71	6	178.9	−0.5	180.0	2.7	13.2	26.8	1.4°S	269.2	60.6	6.6	13.2
Zongshan Fm. (Gamba)	limestone	late Maastricht.	66–69	8	8.9	1.5	6.8	−17.9	10.3	30.1	9.2°S	257.6	52.0	5.5	10.7
Zongshan Fm.) (Total)	limestone	Maastrichtian	65–71	14	4.6	1.1	4.0	−11.2	8.5	22.8	5.7°S	261.4	55.8	4.4	8.6
Jidula Fm. Mbr. II (Gamba)	limestone	late Maastricht. –E. Palaeocene	64–66	3	171.9	−22.2	177.1	−12.9	9.2	180.1	6.5°N	276.2	68.1	4.8	9.4
Jidula Fm. Mbr. III (Gamba)	sandstone	late Maastricht. –E. Palaeocene	63–65	1	354.6	42.6	3.2	18.6	–	–	9.6°N	258.8	71.0	–	–
Lo. Zongpu Fm. (Gamba)	limestone	mid. Palaeocene	58–63	4	170.9	−20.6	176.3	−6.0	12.0	59.3	3.0°N	277.1	64.5	6.0	12.0
Up. Zongpu Fm. (Gamba)	limestone	Late Palaeocene	55–60	6	180.1	−30.2	185.7	−12.4	7.8	74.6	6.3°N	253.7	67.3	4.0	7.9
Zongpu Fm. (Duela)	limestone	mid. Palaeocene	58–63	4	161.8	−3.9	162.1	−1.7	16.5	32.1	0.9°N	304.5	57.9	8.3	16.5
Zongpu Fm. (Total)	limestone	middle–Late Palaeocene	55–63	14	171.9	−20.6	176.2	−7.9	7.5	28.7	4.0°N	277.6	65.4	3.8	7.6

N = number of sites; palaeolatitudes calculated for Gamba (28.3°N, 88.5°E); – = no significant result.

Table 8
Formation mean values for Gamba and Duela calculated from single specimen directions (site A86-I not included)

Stratigraphic unit	Lithology	Geologic age	Numerical age (Ma)	n	In situ		Tilt corrected		α_{95}	k	Palaeo-latitude	North pole position			
					Dec.	Inc.	Dec.	Inc.				Long.	Lat.	dp	dm
Gamba Group (Gamba)	graywacke	mid.–l. Albian	98–107	23	352.6	23.2	352.0	−40.0	7.9	15.7	22.8°S	277.9	38.4	5.7	9.5
Zongshan Fm. (Duela)	limestone	Maastrichtian	65–71	46	179.7	−0.7	180.6	1.2	6.5	11.5	0.6°S	267.3	61.1	3.3	6.5
Zongshan Fm. (Gamba)	limestone	late Maastricht.	66–69	110	4.8	−1.8	5.5	−14.8	4.7	9.1	7.5°S	259.2	53.8	2.5	4.8
Zongshan Fm. (Total)	limestone	Maastrichtian	65–71	156	3.4	−0.9	4.0	−10.7	3.9	9.3	5.4°S	261.4	56.1	2.0	4.0
Jidula Fm. Mbr. II (Gamba)	limestone	late Maastricht. –E. Palaeocene	64–66	19	170.2	−21.9	175.5	−13.1	4.3	62.0	6.6°N	280.5	67.9	2.2	4.4
Jidula Fm. Mbr. III (Gamba)	sandstone	late Maastricht. –E. Palaeocene	63–65	37	354.6	42.6	3.2	18.6	8.0	9.6	9.6°N	258.8	71.0	4.3	8.3
Lo. Zongpu Fm. (Gamba)	limestone	mid. Palaeocene	58–63	42	166.7	−26.2	174.1	−10.2	5.0	20.8	5.1°N	283.2	66.2	2.6	5.1
Up. Zongpu Fm. (Gamba)	limestone	Late Palaeocene	55–60	48	179.5	−30.4	185.3	−11.9	5.2	16.8	6.0°N	254.8	67.2	2.7	5.3
Zongpu Fm. (Duela)	limestone	mid. Palaeocene	58–63	23	161.0	−2.0	161.5	−1.1	7.0	19.6	0.6°N	304.2	57.1	3.5	7.0
Zongpu Fm. (Total)	limestone	middle–Late Palaeocene	55–63	113	170.7	−23.4	176.2	−9.2	3.6	14.9	4.6°N	277.9	66.1	1.8	3.6

n = Number of specimens; palaeolatitudes calculated for Gamba (28.3°N, 88.5°E).

tion at the sampling area does not lie within the α_{95} circle around the ChRM site mean values in situ (with exception of site B86-S, sandstone from the Jidula Formation), and (2) stable PSD/SD magnetite is identified as the predominant carrier of the ChRM.

Several arguments support primary origin for the ChRM: (3) Antiparallel normal and reverse polarities are observed from the Zongshan, Jidula and Zongpu formations. The observed polarities are in agreement with the magnetic polarity time scale (Fig. 14, after Ogg, 1995), namely, normal polarity for the Albian Gamba Group Member II, mixed polarity for the Maastrichtian Zongshan Formation, reverse polarity for the Jidula Member II (deposited around the Cretaceous/Tertiary time boundary), predominantly normal polarity for the subsequent Jidula Member III and nearly exclusively reverse polarity for the middle–Late Palaeocene Zongpu Formation. (4) Palaeolatitudes show a northward progress with decreasing rock age as is expected from the northward drift of India. (5) No signs of metamorphism could be detected in the sampled units. (6) A positive fold test (McElhinny, 1964) is obtained for the Zongshan Formation (members II, III and IV) that is significant at the 99% level with $k_{\text{tilt cor}}/k_{\text{in situ}} = 4.28$ ($N = 14$ sites). Due to the rather uniform bedding dip, the high significance of the fold test depends mainly on site A23 and on its correct polarity interpretation (Table 7; Fig. 13). Site A23 was drilled on an overturned limb of a S-vergent fold. The structure has been well established through detailed tectonic observations (Burda, 1993). Ten out of a total of eleven treated specimens from site A23 yielded a ChRM direction. The in situ ChRM directions obtained from the overturned bedding of site A23 are all reverse, whereas the in situ ChRM directions

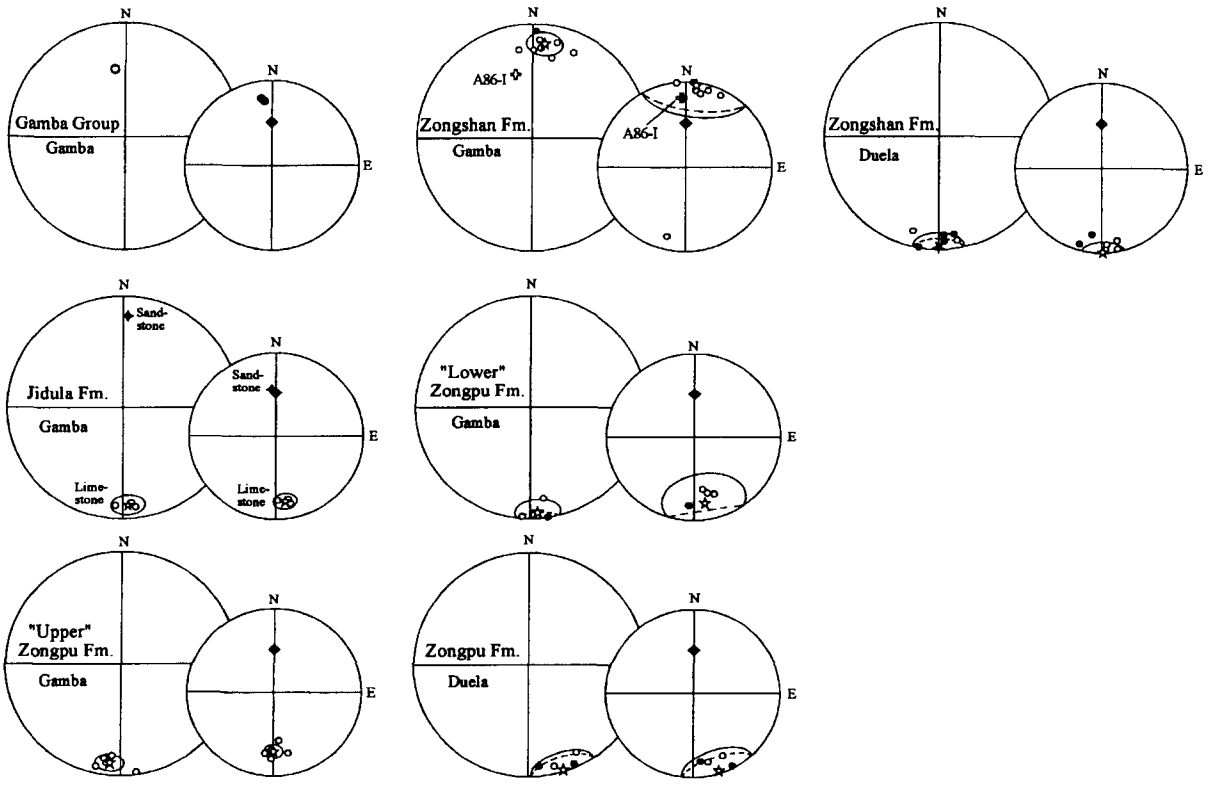


Fig. 13. Equal area projections of site mean directions, divided into eight groups (see also Table 7). The tilt corrected site mean directions are plotted in the larger figures, the in situ site mean directions in the smaller figures. Site mean directions are plotted as circles or dots (except Jidula Member III, star; site A86-I, cross), group mean directions are plotted as stars with corresponding α_{95} circles, and the present dipole field direction is plotted as a rhomb. Open symbols indicate negative inclinations, filled symbols indicate positive inclination.

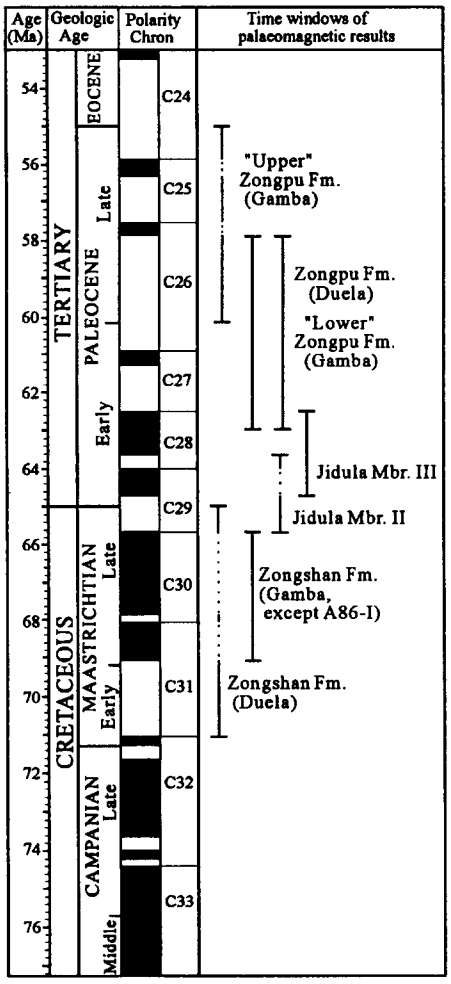


Fig. 14. Magnetic polarity time scale (after Ogg, 1995) plotted together with possible ranges of remanence ages according to rock ages and polarities of ChRM directions.

from all other — normally bedded — specimens from the Zongshan Formation are exclusively of normal polarity. Hence, it seems to be a fairly reliable fact that the observed reverse polarity of the ChRM directions of site A23 is due to the overturned bedding. If site A23 is excluded, the $k_{\text{tilt cor}}/k_{\text{in situ}}$ becomes 0.93, and the fold test is no longer significant. All Zongpu Formation sites show a fold test that is significant at the 95% level ($k_{\text{tilt cor}}/k_{\text{in situ}} = 2.10$, for $N = 14$ sites), but is based on sites with only moderately different bedding (Fig. 13). Fold tests for the Gamba Group and the Jidula Formation are not significant due to the uniform bedding dip and the small number of sites. Partial unfolding tests were not applied, because the data seem not to be sufficient (in terms of different bedding attitudes) and because Tauxe and Watson (1994) showed that possible block rotations around subvertical axes could lead to misinterpretations.

The high drift rate of India of up to 20 cm/yr requires detailed control of the remanence acquisition ages in order to allow a comparison of our data with an APWP for India. Biostratigraphic dating of the sites is very accurate (Willems and Zhang, 1993) and further information is provided by the magnetic polarity observed for the sites (Fig. 13). The ChRM directions of the Zongshan Formation of Gamba all show a normal polarity. Biostratigraphic control places the transition from Campanian to Maastrichtian within the calcareous marls on top of Limestone I. Consequently the magnetization of the Zongshan Formation (members II and III) of Gamba must have been acquired within either Chron 30N or 31N. Magnetostratigraphy alone does not allow to further constrain the Maastrichtian age for the reverse polarity results from the Zongshan Formation at Duela. It is surprising, that the Zongshan ChRM directions from Gamba and Duela show opposing polarities. We are not able to give a simple explanation here. The Late Cretaceous–Palaeocene formations at Duela are in general comparable to those at Gamba, but there are some differences in detail. The uppermost part of the Zongshan Formation at Gamba is made up of calcareous marls with a Rhodolite facies, whereas it is formed by a limestone layer IV (sites D44, D47 and D48) at Duela. The reverse polarity of the corresponding ChRM directions magnetization could have been acquired during either Chron 29R or 30R. Sites D39, D42 and D53 are from the limestone layer II and their magnetization could have been acquired during Chron 31R or 30R. The limestone layer of the Jidula Formation was deposited around the Cretaceous/Tertiary time boundary and shows a reverse polarity for all specimens. Therefore the magnetization was acquired during Chron 28R or 29R. The predominantly normal polarity ChRM of the subsequent Jidula Member III is interpreted to correspond with Chron 28N or 29N. Correlations with the magnetic polarity time scale provide no further information in the case of the various Zongpu

groups (predominantly reverse polarities) and the Albian Gamba Group within the Cretaceous normal polarity Super-Chron C34N.

5. Discussion

In order to reconstruct the northern extent of Greater India prior to collision with southern Asia and to estimate the magnitude of crustal shortening, observed palaeolatitudes are compared with expected palaeolatitudes for the locality of Gamba. Expected palaeolatitude reference curves are calculated from available data for stable India based on the present-day geography (Table 9). Different reference curves result from the Master APWP (REF A) published by Besse and Courtillot (1991), simulated pole positions (REF B) listed in Klootwijk et al. (1991) and palaeomagnetic results from the Ninetyeast Ridge (Klootwijk et al., 1991, 1992). Expected and observed (our

Table 9
Apparent pole positions for India and expected palaeolatitudes for the reference point Gamba (28.3°N, 88.5°E)

Age (Ma)	Pole latitude (°N)	Pole longitude (°E)	A_{95}	Palaeolatitude for ref. point
Master APWP for India (Besse and Courtillot, 1991)				
42.8	72.8	291.9	4.8	12.4
51.7	66.2	283.3	6.0	5.2
62.0	45.6	281.9	8.9	−15.1
68.7	34.2	281.2	6.3	−26.3
80.7	24.3	282.0	4.2	−35.8
88.6	19.8	284.4	3.6	−39.5
98.4	15.3	285.9	7.1	−43.2
112.4	4.9	287.0	5.9	−52.3
India–Africa relative movement, Africa fixed to hotspot frame (listed in Klootwijk et al., 1991) [a]				
42.7	65.4	305.3	3	7.9
46.2	64.5	303.5	3	6.7
50.3	60.1	304.4	3	3.2
52.6	64.4	293.3	3	4.7
54.7	58.3	298.9	3	0.3
56.1	56.5	296.0	3	−2.1
58.6	54.5	292.3	3	−4.7
60.7	52.3	290.8	3	−7.1
63.0	46.8	292.5	3	−11.8
64.3	46.3	290.4	3	−12.8
66.2	41.9	290.1	3	−17.0
68.5	39.9	285.7	3	−20.0
73.5	35.8	283.0	3	−24.5
84.0	27.0	281.2	3	−33.4
Pole positions obtained from ODP cores from the Ninetyeast Ridge (Klootwijk et al., 1991) [a]				
40	66.8	307.4	3	9.5
45	65.1	304.5	3	7.5
50	63.2	302.0	3	5.3
55	61.5	300.0	3	3.3
60	54.0	292.0	3	−5.2
65	44.8	290.4	3	−14.2
Deccan traps (Vandamme et al., 1991)				
65.5	36.9	281.3	2.4	−23.6

[a] A_{95}, confidence limits estimated at about 3°.

data) palaeolatitudes are plotted in Fig. 15 with their confidence limits (*dp* values). The Master APWP results from selected palaeomagnetic data from Africa, Eurasia, North America (all transferred to India) and India using independent 20 myr time windows. The A_{95} values for the so calculated (REF A) Indian pole positions are relatively high for the time period 70–50 Ma, which is not surprising due to the high drift rate of India at that time. The simulated pole positions (REF B) are derived from India–Africa relative movement with Africa fixed to a hot spot frame. They are based on an updated set of magnetic anomaly analyses, similar to that of Patriat and Achache (1984). The error estimate for these pole positions is about $\pm 3°$ (Patriat and Achache, 1984).

Palaeopositions of our data fall significantly to the north of the three sets of expected palaeolatitudes. The difference can be interpreted as the minimum magnitude of crustal shortening between the Tethyan Himalaya and the Indian Shield at the longitude of the sampling sites (88.5°E) since the time of remanence acquisition. The crustal shortening results are:

Zongshan Formation
(mean age 67.7 Ma)

2100 ± 830 km (REF A)
1470 ± 590 km (REF B)

Zongpu Formation
(mean age 59.2 Ma)

1490 ± 790 km (REF A)
1040 ± 540 km (REF B)

These values are calculated using mean directions for all site means from Gamba and Duela together (Table 7). Expected palaeolatitudes at 67.7 and 59.2 Ma are obtained from the reference curves through linear interpolation. Error determination is a delicate problem; here we followed the Gaussian error propa-

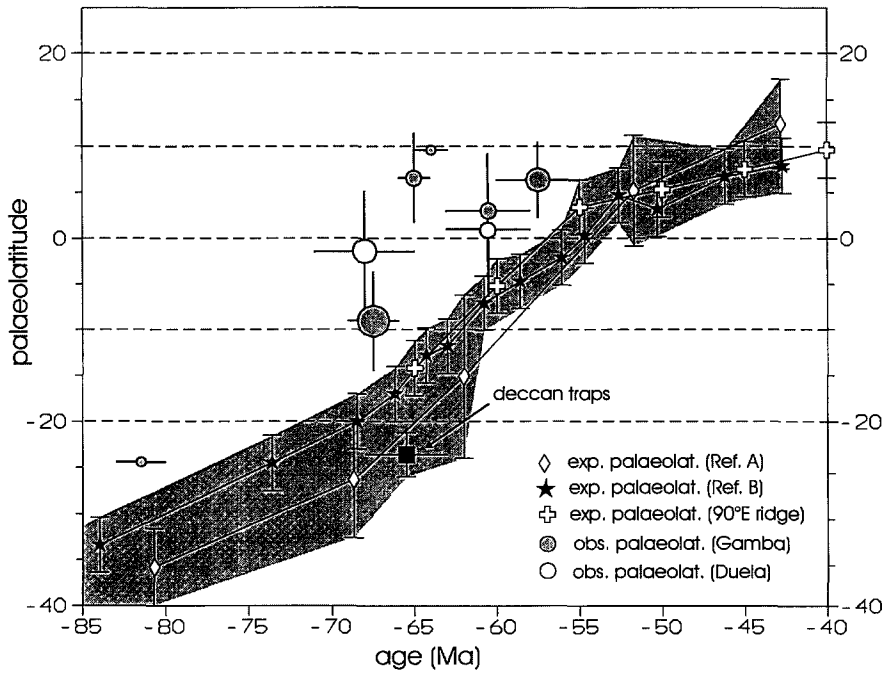


Fig. 15. Expected and observed palaeolatitudes for the Tethyan Himalaya at the reference locality of Gamba (28.3°N, 88.5°E). Data are listed in Table 7 and Table 9. The size of the circles for observed palaeolatitudess is indicative for the number of site mean directions included. Horizontal bars represent the uncertainty in ages, vertical bars represent the confidence limits of the palaeolatitudes. No confidence limits are shown for A86-1 and Jidula Member III, as the results are based on single sites only. The shaded area represents the envelope of the error range for the three types of expected palaeolatitudes.

gation law (errors of the expected palaeolatitudes assumed to be $\pm 6°$ for REF A and $\pm 3°$ for REF B, errors of observed palaeolatitudes are presented as *dp* equivalents to α_{95} values, uncertainties in age not regarded). The Gamba Group and the Jidula Formation are not considered because of the few sites (and specimens) only and an insufficient distribution of site localities. The calculated magnitudes of crustal shortening are valid for pure N–S-directed convergence. They are an underestimate if an E–W-directed component of crustal shortening has occurred, and they are also an underestimate if rotational shortening is considered to contribute to the total magnitude of crustal shortening. The Indian Shield rotated about 20° counterclockwise since the Late Cretaceous, whereas the mean declination of all sites from the Zongshan and Zongpu formations at our reference locality Gamba is about 0°. This difference can be explained by rotational shortening following the oroclinal bending model of Klootwijk et al. (1985). The observed palaeodeclination can be transferred to an orientation parallel to the expected one by recon-

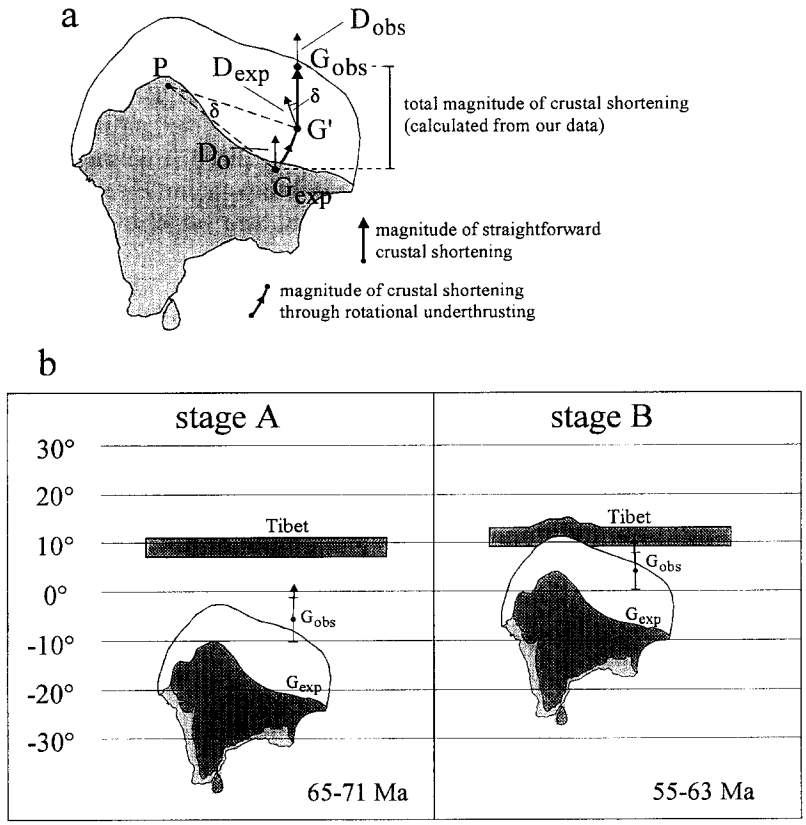


Fig. 16. (a) Rotational and straightforward crustal shortening of the former northern margin of India: rotational shortening at Gamba from G_{exp} to G' (circle around pivot P), straightforward N–S-oriented shortening at Gamba from G' to G_{obs}. G_{exp} represents the expected location of Gamba (assuming that no relative motions between the Tethyan Himalaya and the Indian Shield occurred), G_{obs} represents the observed palaeolatitude (mean of Zongshan and Zongpu formations). D_o is directed to the north, D_{obs} shows the mean palaeodeclination of our results, D_{exp} represents the expected palaeodeclination at Gamba (if the Tethyan Himalaya had followed the counterclockwise rotation of the Indian Shield completely). The angle δ is the difference between D_{obs} and D_{exp}. (b) Reconstruction of the palaeogeographic situation for the late Maastrichtian (*stage A*) and the middle Palaeocene (*stage B*). The dark shaded area represents the continental part of the Indian Plate, the light shaded area represents the shelf. The line north of India shows the former northern extent of Greater India according to our data. G_{exp} and G_{obs} are explained above, however, here G_{obs} is is shown separately for the Zongshan Formation (*stage A*) and the Zongpu Formation (*stage B*). The bars represent the confidence limits of the observed palaeolatitudes, the arrows represent the observed declinations. More details for this figure are given in the text.

struction of the Himalayan arc into a straight line and an additional rotation of this straight segment. The model is not unambiguous, however, for the longitude of Gamba, the path of rotational crustal shortening can be approximated by the arc of circle from G_{exp} to G′ around a pivot at 35°N, 73°E (according to Klootwijk et al., 1985) (Fig. 16a). A magnitude of rotational shortening of about 600 km at the longitude of Gamba results from our data. This value is in agreement with earlier results published by Klootwijk et al. (1985) and Appel et al. (1991). The path G_{exp}–G′ involves a minor E–W-directed component of shortening at the longitude of Gamba (Fig. 16a). Provided that no other E–W-directed component of crustal shortening has occurred, the residual difference of palaeolatitudes between G′ and G_{obs} results in a further straightforward shortening in N–S direction (Fig. 16a), that can be seen as uniform Himalayan-wide shortening. The path G_{exp}–G′–G_{obs} in Fig. 16a defines the minimum magnitude of crustal shortening (approximately 1500 km), but this is certainly based on a simplified model. Of course, straightforward and rotational crustal shortening are likely to have occurred simultaneously.

It is generally accepted that crustal shortening within Greater India occurred by underthrusting (mainly along the MCT) and internal deformation (folding and thrusting). Strong evidence for wholesale underthrusting of Indian crust beneath the Tethyan Himalaya is provided from recent deep seismic reflection profiling beneath southern Tibet (Zhao et al., 1993; Nelson et al., 1995). Palaeomagnetic estimates of the magnitude of crustal shortening between stable India and the Tethyan Himalaya vary between 700 km (Patriat and Achache, 1984), 1000 km (Dewey et al., 1989; Chen et al., 1993b) and a minimum of 1000 km (Klootwijk et al., 1985) for the eastern parts of India. Our result of about 1500 km crustal shortening exceed these values, but is based on an extensive new data set. Of course there remains the problem of error determination, and the lower 95% confidence limit of crustal shortening from our data (about 1000 km) is still in agreement with the other results mentioned above.

The only previously published palaeomagnetic result from Tethyan Himalayan rocks within the period Late Cretaceous–Eocene by Besse et al. (1984) from Tingri (see Fig. 1a) yields a palaeolatitude of 2.5°S $\pm$ 4° for the eastern Tethyan Himalaya at 57 Ma. Crustal shortening on the order of 400 km along the MCT and MBF was inferred by the authors. The result of Besse et al. (1984) is based on primary remanences of three limestone sites (22 specimens only) and one sandstone site (six specimens). Exclusion of the sandstone site result (which seems to be questionable) yields a palaeolatitude of 2.3°S $\pm$ 7.3°, undistinguishable from the original value but with a considerably increased confidence range. Furthermore, palaeomagnetic investigations of Palaeocene and Eocene Tethyan sediments from the same Tingri area (twelve sites with about 130 specimens, mostly limestones), carried out by our own group, show secondary remanences only (Appel et al., submitted). The confidence range of the result of Besse et al. (1984) includes the equator, and inclination shallowing could account for a palaeolatitude too close to the equator. Inclination shallowing can never be ruled out. However, if it occurred to our data from Gamba and Duela the difference between expected and observed palaeolatitudes would increase further.

The magnitude of about 1500 km crustal shortening demands also a discussion of the evolution of the northern margin of Greater India. Recent palaeogeographic reconstructions for Gondwana, like that of Scotese (1991), show an oceanic gap up to about 1000 km between northeastern India and northwestern Australia during Late Jurassic and Early Cretaceous times. This gap disappears, assuming an initially Greater India. An increase of the northern extent of Greater India might have been produced by extension of the Tethyan shelf (crustal thinning) during the rapid north drift, especially when the shelf came close to the subduction zone. Slab pull is indeed believed to act as a dominant force for the movement of the Indian Plate (Patriat and Achache, 1984). The great diversification in the Late Cretaceous–Palaeocene facies of the Tethyan Himalaya in the Zanskar area, with formation of blocks and basins, supports such an extension model for the shelf area as pointed out by Fuchs and Willems (1990). We can not distinguish from our data between models with or without extension, and palaeomagnetic data in general are not very adequate to do that. The strong counterclockwise rotation of the Indian Plate since the separation from Gondwana enlarge the confidence limits far above the resolution needed for

unequivocal conclusions. Nevertheless, comparison of palaeolatitudes calculated from the Gamba Group Member II and site A86-I (Zongshan Formation, Member I) with palaeolatitudes calculated from REF A at least indicates an already existing Greater India in the middle Albian and the early Campanian. The observed palaeopositions for our reference point Gamba are 23.6°/2600 km (Gamba Group) and 11.7°/1300 km (A86-I) north of the expected ones.

A very simple estimate of volume balancing documents that 1500 km of crustal shortening at the northern margin of India are possible in principle. The sediments deposited in Himalayan foreland basins, fans and deltas account for 2×10^7 km^3 of eroded material (Johnson, 1994), derived mainly from the Higher Himalayan area with minor contributions from the Lesser Himalaya, Tibet and Karakoram. Most of these sediments, about 1.25×10^7 km^3, are stored in the Bengal fan (Curray, 1994). Another 2×10^7 km^3 of material is stored in the 70 km thick Himalayan crust (Johnson, 1994). With an initial thinned Indian crust of 20–25 km thickness, as assumed by Dewey et al. (1989), this accounts for a block of about 1100 km length (with an E–W width of 2000 km for the subduction zone). Furthermore, the Indian slab was subducted beneath southern Tibet by a minimal length of 220–400 km (Patriat and Achache, 1984; Harrison et al., 1992; Zhao et al., 1993; Nelson et al., 1995). This leads to a total of up to 1500 km of crustal shortening, without necessarily requiring an unlikely high magnitude of underplating of the Indian crust beneath Tibet or subduction of continental crust.

Implications of our data on the timing of collision of Greater India with Eurasia also require control on the palaeoposition of the southern margin of Eurasia. The Master APWP for Eurasia (Besse and Courtillot, 1991) would yield a Palaeocene palaeolatitude of 28.5°N for the southern margin of stable Eurasia (at the longitude 88.5°E) based on the present-day geography. The southern edge of Eurasia was formed by the Lhasa block at that time. Comparison with palaeolatitude observations from this Lhasa block indicate post-collisional shortening of about 2000 km throughout southern Asia, comparable to the magnitude calculated by Patriat and Achache (1984) on the basis of seafloor spreading data from the Indian ocean. These palaeomagnetic observations from the Cretaceous Takena Formation and the early Tertiary Lingzizong volcanics (Pozzi et al., 1982; Westphal et al., 1983; Westphal and Pozzi, 1983; Achache et al., 1984; Lin and Watts, 1988) yield palaeolatitudes for the Lhasa block ranging from 6 up to 20°N. However, according to Klootwijk et al. (1986, 1991) and Chen et al. (1993a), possible partial overprints limit the reliability of these data. A recently published Late Cretaceous pole (Chen et al., 1993a) indicates a palaeolatitude of 7°N for the southern margin of the Lhasa block. For the middle to Late Palaeocene Zongpu Formation we determined a palaeolatitude of 4°N for the Tethyan Himalaya at Gamba. A palaeogeographic reconstruction is shown in Fig. 16b. The position of stable India results from the mean of REF A and REF B. The northern margin of Greater India is drawn according to the mean value of observed palaeolatitudes from the Zongshan Formation and the Zongpu Formation, taking into account a decreasing magnitude of rotational shortening from east to west (700 km at Gamba, zero at pivot P; see Fig. 16a). At the time of deposition of the Zongshan Formation there is still a wide oceanic gap between Greater India and Eurasia in the east. These data suggest that initial contact of Greater India and Eurasia has occurred close to the Cretaceous/Tertiary time boundary (65–60 Ma). A Late Palaeocene to Early Eocene age (55–50 Ma) is favoured for completion of suturing.

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